THE ALL-CHORDS GRAPH OF A TRIANGLE WITH n POINTS IN GENERAL POSITION ON EACH SIDE HAS $\frac{3}{2}(3n^4+4n^3+3n^2+4n+2)$ EDGES

MATH0

Abstract. Place n points in the relative interior of each side of an equilateral triangle, join every pair of the $3n+3$ boundary points (corners included) by a segment, and assume that no interior point lies on three segments. We prove that the resulting planar graph has $\frac{3}{2}(3n^4+4n^3+3n^2+4n+2)$ edges for every $n \geq 0$ and every such placement, as conjectured by S. R. Shannon and N. J. A. Sloane in OEIS A367119 (2023), by counting the $3n(n+1)$ chords and the $\frac{3}{4}n^2(n+1)(3n+1)$ crossing pairs of chords. The formula also follows in one line from the vertex count that the companion entry A367117 states as a theorem, so we classify the result as the first formal proof of a known result. The proof, including the geometric facts behind the count, is verified in Lean 4 with Mathlib against a statement that math0 formalized from the OEIS entry and froze after a fidelity review, before any proof attempt.

1. Introduction

Place n points in the relative interior of each side of an equilateral triangle Δ and join every pair of the $3n+3$ boundary points (corners included) by a segment. Pairs on a common side give pieces of that side; every other pair gives a *chord* through the interior. If no interior point of Δ lies on three of the segments (general position), the drawing is a planar graph whose vertices are the boundary points and the crossing points of chords. The OEIS entries A367117, A367118 and A367119 [9, 10, 11], contributed by Scott R. Shannon and N. J. A. Sloane on November 5, 2023, count its vertices, regions and edges, and A367119 conjectures that there are $\frac{3}{2}(3n^4+4n^3+3n^2+4n+2)$ edges. We prove this for every placement in general position (Theorem 2), so the count does not depend on the placement, as the entry presupposes.

Prior work and status. A367119 (last revised November 9, 2023) and A367118 still list the edge and region formulas as conjectures, and so does the edge table A367190 [13] for its triangle row. The vertex entry A367117, however, states as a theorem (without an argument) that there are $V(n) = \frac{3}{4}(n+1)(3n^3+n^2+4)$ vertices, the table A367183 [12] lists this row outside its conjectured rows, and a formula of Shannon and Sloane in A213827 [5] (November 9, 2023) equates $n^2(n+1)(3n+1)/4$ with $(V(n)-3(n+1))/3$, a third of the interior crossings (it cites A367177, evidently a typo for A367117). The edge formula follows from $V(n)$ in one line: a chord carrying m crossings is cut into $m+1$ edges, the boundary into $3n+3$ edges, and there are $3n(n+1)$ chords, so $E = 3(n+1)+3n(n+1)+2(V-3n-3) = 2V+3(n^2-1)$. For the variant in which the corners are not endpoints of chords, Sloane proved the crossing count in A365929 [6] (November 7, 2023) and marked the edge and region formulas of A366932 and A367015 [7, 8] as theorems (December 31, 2025); A392261, A392282, A392228 (January 2026) treat k points per side of a regular n-gon, corners excluded, with a proof by Shannon attached to A392261 [14] that we did not consult. Counting crossings of chords in general position by 4-subsets of endpoints is classical; Poonen and Rubinstein [4] treat the non-generic case of the diagonals of a regular polygon. The math0 referees' searches (arXiv, Mathematics Stack Exchange, the web) found no source that explicitly proves the A367119 conjecture, and it is not in formal-conjectures

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[1]. Since the formula is an immediate consequence of the vertex count recorded in A367117, we classify the result as the *first formal proof of a known result*. Starting from the geometric definitions, the Lean proof also covers what the count takes for granted: chords cross exactly when their endpoints interleave, crossing points on a chord are distinct, and the sides carry no vertices besides the boundary points.

2. Statement

Write $[p,q]$ and (p,q) for closed and open segments in $\mathbb{C}=\mathbb{R}^2$. Let $c_0=0$, $c_1=1$, $c_2=\frac{1}{2}+\frac{\sqrt{3}}{2}i$ (indices modulo 3), $\Delta=\operatorname{conv}\{c_0,c_1,c_2\}$, $S_j=[c_j,c_{j+1}]$ and $S_j^\circ=(c_j,c_{j+1})$.

Definition 1. Let $n\ge 0$. A *placement* of n points per side is a finite set $B\subset\mathbb{C}$ with $c_0,c_1,c_2\in B$, $B\subseteq S_0\cup S_1\cup S_2$ and $|B\cap S_j^\circ|=n$ for $j=0,1,2$. The *drawn segments* are the $[p,q]$ with $p\neq q$ in B. B is in *general position* if no point of $\operatorname{int}\Delta$ lies on three pairwise distinct drawn segments. The *vertex set* $V(B)$ consists of the points of B and every z with $s\cap t=\{z\}$ for two drawn segments s,t. In the graph $G(B)$ on $\mathbb{C}$, distinct u,v are adjacent if $u,v\in V(B)$, some drawn segment contains both, and $V(B)\cap(u,v)=\emptyset$. $E(B)$ is the number of edges of $G(B)$.

The entry A367119 [11] has offset 0 and 36 listed terms $3,24,153,588,1635,3708,\dots,7015788$. Its name, first comment and conjecture (Shannon and Sloane, November 5, 2023) read verbatim:

> Place n points in general position on each side of an equilateral triangle, and join every pair of the 3*n+3 boundary points by a chord; sequence gives number of edges in the resulting planar graph.
> "In general position" implies that the internal lines (or chords) only have simple intersections. There is no interior point where three or more chords meet.
> Conjecture: a(n) = (3/2)*(3*n^4 + 4*n^3 + 3*n^2 + 4*n + 2).

Theorem 2. *For every $n\ge 0$ and every placement B of n points per side in general position, $E(B)=\frac{3}{2}\bigl(3n^4+4n^3+3n^2+4n+2\bigr)$.*

Lean formalization. The statement file `lean/Math0/Statements/OEIS/A367119.lean` (SHA-256 `0972bcc5...415bc4`) was written by math0, reviewed for fidelity and frozen before any proof attempt. Its definitions and statement read verbatim (namespace `Math0.Statements.A367119`; docstrings and the computable part omitted):

```
noncomputable def corner : Fin 3 → ℂ := ![0, 1, ⟨1 / 2, Real.sqrt 3 / 2⟩]
noncomputable def triangle : Set ℂ := convexHull ℝ (Set.range corner)
def IsPlacement (n : ℕ) (B : Finset ℂ) : Prop :=
  (∀ j : Fin 3, corner j ∈ B) ∧
  (∀ z ∈ B, ∃ j : Fin 3, z ∈ segment ℝ (corner j) (corner (j + 1))) ∧
  (∀ j : Fin 3, ((B : Set ℂ) ∩ openSegment ℝ (corner j) (corner (j + 1))).ncard = n)
def drawn (B : Finset ℂ) : Set (Set ℂ) :=
  {s | ∃ p ∈ B, ∃ q ∈ B, p ≠ q ∧ s = segment ℝ p q}
def GeneralPosition (B : Finset ℂ) : Prop :=
  ∀ z ∈ interior triangle, ∀ s₁ ∈ drawn B, ∀ s₂ ∈ drawn B, ∀ s₃ ∈ drawn B,
    z ∈ s₁ → z ∈ s₂ → z ∈ s₃ → s₁ = s₂ ∨ s₁ = s₃ ∨ s₂ = s₃
def vertices (B : Finset ℂ) : Set ℂ :=
  (B : Set ℂ) ∪ {z | ∃ s ∈ drawn B, ∃ t ∈ drawn B, s ∩ t = {z}}
def planarGraph (B : Finset ℂ) : SimpleGraph ℂ :=
  SimpleGraph.fromRel fun u v =>
    u ∈ vertices B ∧ v ∈ vertices B ∧ (∃ s ∈ drawn B, u ∈ s ∧ v ∈ s) ∧
      ∀ w ∈ vertices B, w ∉ openSegment ℝ u v
noncomputable def edges (B : Finset ℂ) : ℕ := (planarGraph B).edgeSet.ncard
def statement : Prop :=
  ∀ (n : ℕ) (B : Finset ℂ), IsPlacement n B → GeneralPosition B →
    (edges B : ℚ) = 3 / 2 * (3 * n ^ 4 + 4 * n ^ 3 + 3 * n ^ 2 + 4 * n + 2)
```

Encoding. `corner j` is c_j (`Fin 3` arithmetic makes side 2 run from c_2 to c_0); the other definitions are those of Definition 1. `SimpleGraph.fromRel` symmetrizes the relation and removes loops;

the edge set is finite, so `ncard` has no junk value. General position is the entry's comment read literally (a drawn segment through an interior point is a chord by Lemma 6(e), so it concerns chords only), and $s \cap t = \{z\}$ keeps the points of overlapping pieces of a side out of the vertex set. The conclusion is an identity in $\mathbb{Q}$ for all $n \geq 0$, matching the offset. It does not assert that general placements exist; they do (not formalized): the concurrence of three given chords at an interior point forces a polynomial equation in the positions of the $3n$ non-corner points that is not identically satisfied, since three such chords have six distinct endpoints and moving a non-corner one along its side breaks the concurrence; so a generic placement is in general position.

Data check. The statement file defines a computable model `a`: it numbers the boundary points $0, \dots, 3n+2$ cyclically with corner j at $j(n+1)$, takes as chords the pairs not on a common closed side, lets two chords cross when their endpoints interleave, and sets $\mathtt{a}(n) = (3n+3) + \sum_{\text{chords } c}(1 + \#\{\text{chords crossing } c\})$. The theorem `a_data` checks $\mathtt{a}(0), \dots, \mathtt{a}(10)$ against the first eleven listed terms by `decide +kernel` (axiom: `propext` only). The identity $\mathtt{a}(n) = E(B)$ for placements in general position is not proved in Lean (it follows from Proposition 8 and Lemma 6(c),(d), which are), and `statement` does not use `a`. The formula agrees with all 36 listed terms, and the referees' exact rational simulation of Definition 1 on random placements gave $3, 24, 153, 588, 1635, 3708$ for $n \leq 5$.

3. Proof

We follow `lean/Math0/Frontier/OEIS/A367119.lean`; brackets give the corresponding (private) Lean declarations. For $p, q, z \in \mathbb{C}$ let $D(p,q,z) = \operatorname{Re}(q-p)\operatorname{Im}(z-p) - \operatorname{Im}(q-p)\operatorname{Re}(z-p)$; it is affine in z and vanishes on $[p,q]$ [`D_comb`, `D_of_mem_segment`].

Lemma 3. *Let $p, q, r, s \in \mathbb{C}$. (a)* [`seg_inter_sub`] *If $D(p,q,r) \neq 0$, then $[p,q] \cap [r,s]$ has at most one point. (b)* [`crossPt_mem`] *If $D(p,q,r)D(p,q,s) < 0$ and $D(r,s,p)D(r,s,q) < 0$, then $(p,q) \cap (r,s) \neq \emptyset$. (c)* [`seg_pos`, `seg_neg`] *If $D(p,q,r)$, $D(p,q,s)$ are both positive or both negative, $D(p,q,\cdot)$ has no zero on $[r,s]$.*

Proof. (a) $(1-\lambda)r + \lambda s \in [p,q]$ forces $(1-\lambda)D(p,q,r) + \lambda D(p,q,s) = 0$, an affine equation in λ that is not identically zero. (b) With $\alpha = D(p,q,r)$, $\beta = D(p,q,s)$, the point $X = (\alpha s - \beta r)/(\alpha - \beta)$ has positive coefficients on r, s, and a direct computation using $D(r,s,q) - D(r,s,p) = \alpha - \beta$ writes it with the positive coefficients $D(r,s,q)/(\alpha-\beta)$, $-D(r,s,p)/(\alpha-\beta)$ on p, q. (c) On $[r,s]$, $D(p,q,\cdot)$ takes convex combinations of its values at r and s. □

For $t \in [0,3)$ let $P(t) = x(t) + i\sqrt{3}\,y(t)$, where $(x,y)(t)$ is $(t,0)$ for $t \leq 1$, $\left(\frac{3-t}{2}, \frac{t-1}{2}\right)$ for $1 < t \leq 2$ and $\left(\frac{3-t}{2}, \frac{3-t}{2}\right)$ for $2 < t < 3$; so P runs along S_0, S_1, S_2 in turn and $P(j) = c_j$. Put $\Sigma_0 = [0,1]$, $\Sigma_1 = [1,2]$, $\Sigma_2 = [2,3) \cup \{0\}$ and $I_0 = (0,1)$, $I_1 = (1,2)$, $I_2 = (2,3)$.

Lemma 4. *Let $t \in [0,3)$. (a)* [`P_inj`] *P is injective on $[0,3)$, with left inverse $\theta(z) = \operatorname{Re} z$ if $\operatorname{Im} z = 0$ and $\theta(z) = 3 - 2\operatorname{Re} z$ otherwise. (b)* [`P_mem_side_iff`, `P_mem_openSide_iff`] *$P(t) \in S_j \iff t \in \Sigma_j$, $P(t) \in S_j^\circ \iff t \in I_j$, and $S_j = P(\Sigma_j)$. (c)* [`Dline_pos_iff`] *$D(c_0, c_1, P(t)) = \sqrt{3}\,y$, $D(c_1, c_2, P(t)) = \frac{\sqrt{3}}{2}(1-x-y)$, $D(c_2, c_0, P(t)) = \frac{\sqrt{3}}{2}(x-y)$. All three are ≥ 0, and $D(c_j, c_{j+1}, P(t)) = 0$ if and only if $t \in \Sigma_j$. (d)* [`mem_triangle_of`, `mem_interior_of`] *If $D(c_j, c_{j+1}, z) \geq 0$ for $j = 0, 1, 2$, then $z \in \Delta$; if all three are > 0, then $z \in \operatorname{int}\Delta$. (e)* [`exists_T`] *If B is a placement of n points per side and $T = \theta(B)$, then $T \subset [0,3)$, $B = P(T)$, $0, 1, 2 \in T$, $|T \cap I_j| = n$, $|T| = 3n+3$ and $|T \cap \Sigma_j| = n+2$.*

Proof. (a)–(c) are computations; $\operatorname{Im} P(t) = 0$ exactly when $t \leq 1$, and $\operatorname{Re} P(t) = (3-t)/2$ for $t > 1$. (d) The numbers $\frac{2}{\sqrt{3}}\big(D(c_1,c_2,z), D(c_2,c_0,z), D(c_0,c_1,z)\big)$ sum to 1 and are the barycentric coordinates of z; the set where all are positive is open and lies in Δ. (e) By (b)

each $z \in B$ is $P(t)$ with $t = \theta(z) \in [0,3)$; the corners give $0,1,2 \in T$; $B \cap S_j^\circ = P(T \cap I_j)$ and (a) give $|T \cap I_j| = n$; and $T = \{0,1,2\} \sqcup \bigsqcup_j (T \cap I_j)$, while $T \cap \Sigma_j$ is $T \cap I_j$ plus two corner parameters. □

From now on B is a placement in general position and $T = \theta(B)$. A *chord* is a pair (a,b) with $a < b$ in T such that no Σ_j contains both a and b; its segment is $[P(a), P(b)]$. Let $E(a,b,c) = D(P(a), P(b), P(c))/\sqrt{3}$ [D_P].

Lemma 5. *(a)* [E_pos] *If $0 \le a < b < c < 3$ and no Σ_j contains all of a,b,c, then $E(a,b,c) > 0$. (b)* [E_sign, P_not_mem_seg] *If (a,b) is a chord and $t \in [0,3) \setminus \{a,b\}$, then $D(P(a), P(b), P(t))$ is negative if $a < t < b$ and positive otherwise. So no other point of B lies on the chord.*

Proof. (a) Let $A = [0,1]$, $B' = (1,2]$, $C = (2,3)$, and write XYZ for the case $a \in X$, $b \in Y$, $c \in Z$. In the cases AAA, $B'B'B'$, CCC the parameters lie in one Σ_j; otherwise $2E(a,b,c)$ is

$$\begin{aligned} AAB'&: (b-a)(c-1), & AAC&: (b-a)(3-c),\\ AB'B'&: (1-a)(c-b), & B'B'C&: (b-a)(c-2),\\ ACC&: a(c-b), & B'CC&: (2-a)(c-b),\\ AB'C&: (1-a)(2-b)(3-c) + a(b-1)(c-2). \end{aligned}$$

All factors are ≥ 0. A zero value occurs only for $AB'B'$ with $a = 1$, ACC with $a = 0$, $B'CC$ with $a = 2$, or $AB'C$ with $a = 0$, $b = 2$, and in each of these cases a, b, c lie in one Σ_j. (Lean: `nlinarith` in each case.) (b) The triples (a,t,b), (t,a,b), (a,b,t) contain a, b, so no Σ_j contains them. Apply (a) to the one in increasing order and use $E(a,b,t) = -E(a,t,b) = E(t,a,b)$. □

Chords (a,b) and (e,f) *interleave* if $a < e < b < f$ or $e < a < f < b$ [crossR].

Lemma 6. *Let $c = (a,b)$ and $d = (e,f)$ be chords. (a)* [oseg_interior, oseg_not_side] *$(P(a), P(b)) \subset \operatorname{int} \Delta$, and $(P(a), P(b))$ misses $S_0 \cup S_1 \cup S_2$. (b)* [seg_inter_unique] *If $c \ne d$, the two chord segments have at most one common point. (c)* [X_mem, seg_inter_X] *If c, d interleave, their segments meet in exactly one point $X(c,d)$, which lies in both open segments. (d)* [crossR_of_mem] *If $c \ne d$ and $(P(a), P(b)) \cap [P(e), P(f)] \ne \emptyset$, then c and d interleave. (e)* [classify] *Every drawn segment is $[P(e'), P(f')]$ with $e' \ne f'$ in T; either it is a chord segment, or $e', f' \in \Sigma_j$ for some j and it lies in S_j. (f)* [Sd_inter, side_inter_corner] *For $i \ne j$, $\Sigma_i \cap \Sigma_j$ is one corner parameter and $S_i \cap S_j$ is one corner. (g)* [oneD] *Two nondegenerate subsegments of a segment $[p,q]$, $p \ne q$, whose open segments share a point z, share another point.*

Proof. (a) By Lemma 4(c), $D(c_j, c_{j+1}, \cdot)$ is ≥ 0 at $P(a), P(b)$ and > 0 at one of them, as a, b are not both in Σ_j; so it is > 0 on $(P(a), P(b))$. Apply Lemma 4(d); the sides lie on the zero lines. (b) Some endpoint of d is not an endpoint of c; by Lemma 5(b) it lies off the line of c, and Lemma 3(a) applies. (c) By Lemma 5(b), interleaving puts $P(e), P(f)$ strictly on opposite sides of the line of c, and $P(a), P(b)$ on opposite sides of the line of d. Apply Lemma 3(b), then (b). (d) Let z be a common point. If c, d share an endpoint $P(u)$, the other endpoint of d is off the line of c, so $z = P(u)$ by Lemma 3(a), but $P(u) \notin (P(a), P(b))$. If they share no endpoint and do not interleave, then e, f both lie in (a,b) or both outside $[a,b]$, so $D(P(a), P(b), \cdot)$ has the same strict sign at $P(e)$, $P(f)$ (Lemma 5(b)) and no zero on $[P(e), P(f)]$ (Lemma 3(c)), contradicting $D(P(a), P(b), z) = 0$. (e) holds as the sides are convex, and (f) follows from Lemma 4(b). (g) Pull back along $\lambda \mapsto p + \lambda(q-p)$: two real intervals whose interiors contain ζ share $[\zeta, \zeta + \varepsilon]$ for some $\varepsilon > 0$. □

For a chord c let $\kappa(c)$ be the number of chords interleaving with c.

Proposition 7. *(a)* [V_chord, VC_card] *For a chord $c = (a,b)$, $V(B) \cap [P(a), P(b)] = \{P(a), P(b)\} \cup \{X(c,d) : d \text{ interleaves } c\}$, a set of exactly $2 + \kappa(c)$ points. (b)* [V_side, VS_card] *$V(B) \cap S_j = P(T \cap \Sigma_j)$, a set of $n+2$ points.*

Proof. (a) The endpoints are in B, and $X(c,d)$ is a vertex since $\{X(c,d)\} = [P(a),P(b)] \cap [P(e),P(f)]$. Conversely, let $z \in V(B) \cap [P(a),P(b)]$, $z \neq P(a), P(b)$. Then $z \in (P(a),P(b)) \subset \operatorname{int}\Delta$, and $z \notin B$ by Lemma 5(b), so $\{z\} = s \cap t$ for drawn segments $s \neq t$ (a drawn segment has two points). General position at z, applied to $[P(a),P(b)]$, s, t, shows that one of s,t is $[P(a),P(b)]$; the other contains z, so by Lemma 6(e),(a) it is the segment of a chord $d \neq c$, and by Lemma 6(d),(c) d interleaves c and $z = X(c,d)$. The points $X(c,d)$ lie in the open segment, so they differ from the endpoints, and $P(a) \neq P(b)$. If $X(c,d) = X(c,d')$ with $d \neq d'$, this interior point lies on the segments of c, d, d', which are pairwise distinct by Lemma 6(b) [`seg_ne`], contradicting general position. (b) The points of B on S_j are vertices and form $P(T \cap \Sigma_j)$, of size $n+2$. Conversely, let $z \in V(B) \cap S_j$, $z \notin B$, so $\{z\} = s \cap t$ for drawn segments s,t and z lies in the open segment of each. By Lemma 6(a),(e), $s \subseteq S_k$ and $t \subseteq S_{k'}$. If $k = k'$, Lemma 6(g) gives a second point of $s \cap t$; if $k \neq k'$, then z is a corner by Lemma 6(f), so $z \in B$. Both are impossible. □

Proposition 8 (`edgeSet_eq`, `EF_card`, `edges_eq`)**.** *Call distinct u, v in a finite set $F \subset \mathbb{C}$* consecutive *if $F \cap (u,v) = \emptyset$. The edge set of $G(B)$ is the disjoint union, over the sides S_j and the chord segments M, of the sets of consecutive pairs of $V(B) \cap S_j$ and of $V(B) \cap M$. Hence*

$$E(B) = \sum_{j=0}^{2} \big((n+2) - 1\big) + \sum_{c} \big(\kappa(c) + 1\big) = 3(n+1) + \#\{\textit{chords}\} + \sum_{c} \kappa(c).$$

Proof. Let M be a side or a chord segment and $F_M = V(B) \cap M$. As M is convex, a pair $\{u,v\} \subseteq M$ is an edge exactly when it is a consecutive pair of F_M [`consF_iff`]. Every edge lies on a drawn segment, hence on some M (Lemma 6(e)), and every M is drawn. The families are disjoint since an edge has two points: distinct chord segments share at most one point (Lemma 6(b)), two sides share one corner (Lemma 6(f)), and S_j meets the segment of a chord (a,b) only in its endpoints (Lemma 6(a)), which are not both on S_j. A nonempty finite subset of a nondegenerate segment, pulled back to reals $\lambda_1 < \cdots < \lambda_k$ along an affine parametrization, has exactly the $k-1$ consecutive pairs $\{\lambda_i, \lambda_{i+1}\}$ [`consF_card`, `consR_card`]. Proposition 7 gives the sizes. □

Lemma 9. *(a)* [`chords_card`] *There are exactly $3n(n+1)$ chords. (b)* [`diag_iff`] *Let $0 \le w < x < y < z < 3$. For each j, at least three of w,x,y,z lie in Σ_j if and only if $\{w,y\} \subseteq \Sigma_j$ or $\{x,z\} \subseteq \Sigma_j$. Hence (w,y) and (x,z) are both chords if and only if no Σ_j contains three of the four points. (c)* [`crossings_formula`] *$\sum_c \kappa(c) = \frac{3}{2}\, n^2(n+1)(3n+1)$.*

Proof. (a) Of the $(3n+3)(3n+2)$ ordered pairs of distinct parameters, exactly $3(n+2)(n+1)$ lie in a common Σ_j, since no such pair lies in two of them (Lemma 6(f)). Each chord accounts for two of the others, so there are $\frac{1}{2}\big[(3n+3)(3n+2) - 3(n+2)(n+1)\big] = 3n(n+1)$ chords. (b) Every 3-subset of $\{w,x,y,z\}$ contains $\{w,y\}$ or $\{x,z\}$. Conversely, Σ_0, Σ_1 are intervals, so $w,y \in \Sigma_j$ gives $x \in \Sigma_j$ and $x,z \in \Sigma_j$ gives $y \in \Sigma_j$. For $\Sigma_2 = [2,3) \cup \{0\}$: if $w,y \in \Sigma_2$, then $x \in \Sigma_2$ when $w \ge 2$, and $z > y \ge 2$ when $w = 0$; if $x,z \in \Sigma_2$, then $x > 0$ forces $x \ge 2$, so $y \in \Sigma_2$. (In Lean these checks go through a monotone region index $\rho : [0,3) \to \{0,\dots,5\}$ and `omega`.) (c) Interleaving is symmetric and irreflexive, so $\sum_c \kappa(c) = 2N$, where N counts the pairs of chords $(w,y),(x,z)$ with $w < x < y < z$. Sending such a pair to $\{w,x,y,z\}$ is a bijection [`crossH_card`] onto the set $\mathcal{G}$ of 4-subsets of T with no three points in one Σ_j; the inverse sorts the 4-set (`Finset.orderEmbOfFin`) and takes its two diagonals, which are chords by (b). As $|T \cap \Sigma_j| = n+2$ and $|T \setminus \Sigma_j| = 2n+1$, there are $(2n+1)\binom{n+2}{3} + \binom{n+2}{4} = \frac{1}{8}n(n+1)(n+2)(3n+1)$ 4-subsets with at least three points in Σ_j [`bad_card`], and none has three points in each of two $\Sigma_i \neq \Sigma_j$, as $|\Sigma_i \cap \Sigma_j| = 1$ [`not_both_bad`]. By inclusion–exclusion

and $\binom{3n+3}{4} = \frac{3}{8}n(n+1)(3n+1)(3n+2)$,

$$|\mathcal{G}| = \binom{3n+3}{4} - 3\Big[(2n+1)\binom{n+2}{3} + \binom{n+2}{4}\Big] = \tfrac{3}{8}n(n+1)(3n+1)\cdot 2n = \tfrac{3}{4}n^2(n+1)(3n+1).$$

In Lean the binomial coefficients are expanded as rational polynomials and the identity is checked by `ring`. □

Proof of Theorem 2. [`result`] By Proposition 8 and Lemma 9(a),(c),

$$E(B) = 3(n+1) + 3n(n+1) + \tfrac{3}{2}n^2(n+1)(3n+1) = \tfrac{3}{2}\big(3n^4 + 4n^3 + 3n^2 + 4n + 2\big). \qquad \square$$

Remark 10. The same lemmas give (not formalized) $V = 3n+3+\frac{3}{4}n^2(n+1)(3n+1)$ (A367117) and, by Euler's formula, $E - V + 1 = \frac{1}{4}(9n^4 + 12n^3 + 15n^2 + 12n + 4)$ regions (A367118).

4. Machine verification

The proof is the file `lean/Math0/Frontier/OEIS/A367119.lean` of the math0 repository, `https://github.com/Foundation-0/math0`, which imports the frozen statement file and `lean/Math0/Audit.lean`. The main theorem is `Math0.Frontier.A367119.result`, of type `Math0.Statements.A367119.statement`; the auxiliary declarations are private. The file ends with

```
#math0_settles Math0.Frontier.A367119.result Math0.Statements.A367119.statement
```

This command (defined in `lean/Math0/Audit.lean`) fails unless `statement` is a universe-monomorphic `Prop`-valued definition, `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound` (excluding `sorryAx`, `native_decide` and custom axioms), and the type of `result` is syntactically `statement` or its negation, without unfolding. Here `lake env lean` on the file prints no warnings and reports

```
math0 audit: 'Math0.Statements.A367119.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A367119.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A367119.result' proves 'Math0.Statements.A367119.statement'
```

The proof file contains no `sorry`, `native_decide`, `axiom` or `set_option`. We use Lean 4 [2] and Mathlib [3], both `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root; it fetches the Mathlib cache, builds all math0 results with `lake build` (re-running every audit) and checks that the pinned dependencies are unmodified. This result alone: `lake build Math0.Frontier.OEIS.A367119` in `lean`.

References

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